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Beyond grammar checking: the impact of LLM-based instruction on writing strategies for self-regulated learning in EFL contexts

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ABSTRACT

This mixed-methods study examined the impact of integrating large language models (LLMs), specifically ChatGPT, into college English writing instruction on the development of writing strategies for self-regulated learning (WSSRL) among Chinese English as a Foreign Language (EFL) learners. The study adopted a 12-week quasi-experimental design involving two intact classes of first-year undergraduates. The experimental group received LLM-supported instruction, while the control group followed traditional teacher-led instruction. Quantitative data were collected through pre- and post-intervention questionnaires, and qualitative insights were obtained from semi-structured interviews conducted after intervention. The findings showed that LLM-supported instruction enhanced students' use of cognitive, metacognitive and motivational strategies, while also fostering new feedback-related behaviors, including self-initiated feedback seeking and critical engagement with feedback. Although students recognize the immediacy and accessibility of LLM support, they consistently emphasize the irreplaceable role of teacher guidance, particularly in areas requiring contextual judgment and personalized feedback. These findings highlight the potential of LLM-integrated instruction to promote writing strategy use and feedback literacy, and underscore the importance of a balanced collaboration between teachers and LLM in EFL writing instruction.

IMPACT STATEMENT

This study demonstrates that LLM-based writing instruction can move beyond surface-level grammar correction to support EFL learners' development of self-regulated writing strategies. By integrating LLMs into classroom writing instruction, teachers can scaffold learners' planning, monitoring, and revision processes more effectively. The findings offer practical implications for EFL educators who seek to implement AI tools in learner-centered ways.

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1. Introduction

The integration of large language models (LLMs), such as ChatGPT and ERNIE Bot, into English as a Foreign Language (EFL) writing instruction is transforming how EFL learners engage with writing tasks (Huang & Guo, 2024; Xu & Wang, 2025). Unlike earlier computer-assisted language learning tools, LLMs offer multifaceted support, including idea generation (Chang & Li, 2025), text organization (Marzuki et al., 2023) and real-time feedback (Kohnke et al., 2023). This technological advancement reshapes the learner's role from passive recipient to active participant in the writing process. In the Chinese tertiary EFL context, college students generally lack writing autonomy (Xie & Jiang, 2025) and strategic awareness (Guo & Qin, 2010). The introduction of LLMs offers new opportunities to enhance their learning autonomy and strategy awareness. By enabling interactive feedback cycles, LLMs may particularly contribute to the development of students' feedback literacy (Wu, Lee, et al., 2024; Wu, Li, et al., 2024),

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which encompasses the understanding, capacities and dispositions needed to interpret and implement feedback (Carless & Boud, 2018). As advanced tools that combine natural language processing and deep learning, LLMs demonstrate strong potential in supporting personalized learning (Razafinirina et al., 2024) and formative evaluation (Cohn et al., 2024), thereby drawing increasing attention from both educators and researchers.

While previous studies have mainly examined self-regulated learning (SRL) strategies in traditional classroom contexts, recent developments in LLMs have introduced new possibilities for writing instruction. SRL strategy refers to learners' conscious mobilization and maintenance of their cognition, motivation, behavior and emotion in pursuit of specific learning goals (Schunk & Zimmerman, 2006), which are vital in cognitively demanding tasks such as writing (Nückles et al., 2009; Zimmerman, 2002). Writing strategies for SRL (WSSRL) in L2 contexts refer to purposeful and goal-oriented actions that learners adopt to enhance the writing process by making it more effective, manageable and engaging (Teng & Zhang, 2016).

Empirical studies have confirmed the effectiveness of WSSRL-based training in enhancing writing performance (Huang & Zhang, 2022; Wischgoll, 2016), yet such interventions typically rely on teacher guidance or peer interaction. Emerging evidence suggests that LLMs can serve as supportive tools for goal setting, planning and self-monitoring (Liu, Hwang, et al., 2024; Stojanov, 2023), but most research to date focuses on general language skills. In the context of Chinese tertiary EFL writing, little is known about the practical effects of LLM-supported instruction on learners' use of WSSRL. Therefore, this study aims to examine whether integrating LLMs into writing instruction can foster the development of WSSRL among Chinese university EFL learners, thereby addressing the research gap in LLM-assisted writing instruction on learners' use of WSSRL. To this end, this study seeks to address the following research questions:

1. To what extent does LLM-assisted writing instruction enhance EFL learners' WSSRL use compared to conventional instruction?
2. How do EFL learners perceive the impact of LLM-assisted writing instruction on their WSSRL use?

2. Literature review

2.1. Writing strategies for self-regulated learning

SRL has long been a central construct in education, particularly within educational psychology. Zimmerman (2000) defines SRL as a complex process in which learners autonomously generate and apply specific patterns of thinking, feeling and behaving based on particular learning tasks and instructional goals. SRL emphasizes learners' active participation and regulation of their cognitive, emotional, and motivational aspects at various learning stages to achieve learning objectives, which reflects a high degree of learner autonomy. At the strategic level, Pintrich (2004) proposes a conceptual framework for classroom SRL, identifying four major areas for regulation: cognition, motivation/affect, behavior and context.

In L2 writing contexts, researchers generally view WSSRL as an active, constructive, and strategic process in which learners achieve predetermined writing goals through effective monitoring and regulation of their perceptions, motivations and behaviors (Bembenutty, 2011; Zimmerman, 2011). Teng and Zhang (2016) four-dimensional model has been widely recognized as a classic and versatile framework for conceptualizing WSSRL. This model comprises four key dimensions: cognitive, metacognitive, social behavioral and motivational regulation strategies. Cognitive strategies refer to skills students use to process information or knowledge when completing a task (Pintrich et al., 1991). Metacognitive strategies refer to skills used to control and regulate learners' own cognition and the cognitive resources they can apply to meet the demands of particular tasks (Winne, 2011). Social behavioral strategies refer to learners' efforts to regulate their writing through social interaction and the utilization of external resources (Teng, 2024a, 2024b, 2024c). Motivational regulation strategies are learners' deliberate efforts to regulate their writing behaviors by leveraging social and contextual resources (Teng & Zhang, 2018). These four major

Table 1. Structure of WSSRL (Teng & Zhang, 2016).

Dimension	Strategies	Definition
Cognitive strategies	Text processing	Applying linguistic, rhetorical and discourse knowledge to construct coherent and logically organized written texts.
	Course memory	Recalling and utilizing writing-related knowledge (e.g. vocabulary, expressions) learned in previous instruction.
Metacognitive strategies	Idea planning	Generating and organizing ideas before or during writing, including brainstorming and searching for supporting information.
	Goal-oriented Monitoring and evaluation	Setting writing goals, monitoring progress toward objectives, and evaluating mastery of knowledge and writing performance.
Social Behavioral Strategies	Peer learning	Seeking assistance from peers or engaging in collaborative activities to enhance writing quality and overcome challenges.
	Feedback handling	Interpreting, accepting and implementing teacher and peer feedback to improve text quality and writing skills.
Motivational regulation strategies	Interest enhancement	Making writing tasks more engaging and personally relevant to sustain learners' interest and effort.
	Motivational self-talk	Using self-encouragement techniques to maintain motivation and emphasize the importance of writing improvement.
	Emotion control	Managing negative emotions and reducing distractions to maintain focus and persistence during writing.

dimensions encompass a range of specific WSSRL, with detailed definitions of these strategies and their subcategories presented in Table 1.

Existing research consistently underscores the positive impact of WSSRL on writing development. Huang and Zhang (2022) highlight the crucial role of metacognitive strategies (such as planning, monitoring and evaluation) in enhancing learners' writing competence. Empirical evidence also supports the effectiveness of SRL-based intervention: Shen and Bai (2024) find that systematic WSSRL-based training led to significant gains in both English writing performance and strategic use among Chinese university students. Similarly, Zhou and Hiver (2022) report that WSSRL can effectively predict students' engagement in L2 writing classes. Consequently, how to instruct students to use writing strategies effectively has become a focus of current research and instructional practice (Nückles et al., 2009).

Despite growing theoretical clarity, empirical studies on SRL in L2 learning have largely focused on human-facilitated strategy training. For example, Teng and Zhang (2020) note that Chinese university students who received SRL strategy-based instruction outperformed their counterparts in writing performance and showed significant improvements in metacognitive, social behavioral and motivational regulation strategies. Similarly, other studies have found that WSSRL-oriented instruction enhances writing confidence and self-efficacy (Chen, 2022; Shen & Bai, 2024). In recent years, research has begun to incorporate technology into WSSRL-based instruction, highlighting its potential to foster strategic learning behaviors. Empirical evidence suggests that technology-supported WSSRL interventions can improve writing competence while promoting strategy use and motivation. For instance, integrating online resources and digital tools into WSSRL-based instruction has improved students' writing outcomes, self-regulation abilities and acceptance of technology (Tran & Ma, 2025; Yabukoshi & Mizumoto, 2024). However, despite these promising trends, there remains limited empirical research on how advanced technologies, especially LLMs, promote the development of WSSRL in writing.

2.2. Large language models in second language writing

The rapid development of educational technology has created opportunities for integrating technology into classroom teaching (Hyland & Hyland, 2006). Today, artificial intelligence (AI) is bringing profound changes to language education. One type of AI, called LLMs, can generate text that resembles human writing. A common application of LLMs is the 'chatbot', a software program that can simulate human

conversation, such as ChatGPT (Asadi et al., 2025). ChatGPT can produce human-like text based on user input, making it a potentially valuable tool for supporting students' writing practice. Its natural language capabilities can engage and motivate learners (Baskara, 2023). In the learning process, LLMs can be viewed as a 'more knowledgeable other', helping learners construct knowledge and engage in self-regulated writing behaviors through interactive dialogue within their zone of proximal development (Stojanov, 2023). Accordingly, LLMs serve not only as text generation tools (Al Faraby et al., 2024) but also as important mediators for promoting learners' cognitive engagement, dialogic learning and self-regulation (Huang et al., 2023; Lan & Zhou, 2025; Rospigliosi, 2023).

Existing research demonstrates that LLMs offer multidimensional advantages in writing instruction, aligning closely with the three cyclical phases of SRL: forethought, performance and self-reflection (Zimmerman, 2002). First, during the planning stage, corresponding to the forethought phase of SRL, LLMs assist learners in generating ideas, providing examples and organizing frameworks (Baskara, 2023; Meniado et al., 2024), thus enhancing task initiation efficiency and confidence. During the drafting stage, matching the performance phase, LLMs provide learners with comprehensive linguistic support, including vocabulary, grammar, textual coherence and rhetorical strategies (Banihashem et al., 2024; Jiang et al., 2023). Finally, in the revision phase, equivalent to the self-reflection phase, learners often utilize LLMs for language refinement, logical adjustment and stylistic improvement (Kim et al., 2023). The real-time feedback feature of LLMs helps overcome the limitations of delayed feedback in traditional classrooms, enabling students to engage in multiple rounds of revisions during the writing process (Chan & Colloton, 2024). Through real-time feedback, learners can promptly correct grammatical errors, adjust sentence structures and optimize expressions. Empirical research indicates that learners who integrate ChatGPT feedback surpass those in traditional teaching models in terms of reduced grammatical errors and improved overall language accuracy (Jamshed et al., 2024). Furthermore, Asadi et al. (2025) found that combining ChatGPT feedback with teacher feedback significantly improved writing quality, demonstrating the transformative potential of LLMs in personalized support.

However, these advantages also bring potential risks: some learners may mechanically accept LLM-generated outputs, neglecting critical judgment of text content, and even facing issues of plagiarism or over-reliance on AI assistance (Asadi et al., 2025; Teng, 2024a, 2024b, 2024c). To address this challenge, researchers have emphasized the importance of cultivating feedback literacy in LLM-assisted writing environments, which refers to learners' ability to understand, evaluate and effectively apply feedback (Carless & Boud, 2018; Wu, Lee, et al., 2024; Wu, Li, et al., 2024). Unlike traditional feedback from teachers and peers, feedback in LLM-based contexts is characterized as timely, targeted, adaptive and useful (Steiss et al., 2024). This requires learners not only to identify effective feedback but also to make critical choices between LLM-generated suggestions and human feedback (Zhan & Yan, 2025), further underscoring the importance of guiding learners toward strategic use of LLM tools.

Nevertheless, existing research primarily focuses on the impact of LLMs on linguistic accuracy or overall writing quality, with limited empirical evidence on their role in promoting self-regulated writing strategies. Notably, within the Chinese tertiary EFL context, little is known about how LLM integration influences the development of WSSRL across various dimensions and specific strategies. Moreover, most studies rely on automated scoring, which does not capture process-level strategy use during the drafting and revision phases of writing. This methodological gap hinders the understanding of how immediate, conversational LLM feedback interacts with learners' metacognitive planning and feedback-handling behaviors. Therefore, this study aims to address this gap by using a mixed-methods approach to examine learners' strategy use.

3. Methods

3.1. Research design

This study adopted a mixed-methods research design that integrated both quantitative and qualitative approaches to comprehensively examine the effectiveness of LLM-supported writing instruction. The quantitative component involved two naturally occurring classes, and the grouping followed existing class structures, as random assignment was not possible due to institutional scheduling constraints. The

qualitative component utilized semi-structured interviews to provide deeper insights into participants' experiences and emergent behaviors not captured by quantitative measures.

To assess the effectiveness of the intervention, both groups completed the same set of measurements before and after the 12-week instructional period. These included a writing task and a WSSRL questionnaire. In addition to quantitative instruments, qualitative interview data were collected from several participants in the experimental group to gain deeper insights into their learning experiences. The research design was considered suitable based on both ethical and pedagogical factors. Both groups received the same amount of teaching time and support during the semester.

3.2. Participants and context

This study was conducted in the fall semester of 2024 at a university in Beijing. The participants were two parallel classes of non-English major freshmen from the Department of Mechanical Engineering and the Department of Mathematics, with a total of 66 participants. All students were assigned to the highest level (Level A) English courses based on their English scores in the college entrance examination, indicating relatively consistent language proficiency levels.

The experimental group consisted of Mechanical Engineering students ($n = 33$), aged from 17 to 19 years ($M = 18.06$, $SD = 0.43$), with males accounting for 72.7% ($n = 24$). The control group comprised Mathematics majors ($n = 33$), aged between 18 and 19 years old ($M = 18.15$, $SD = 0.36$), with a male proportion of 78.7% ($n = 26$). To ensure the baseline comparability between groups, a chi-square test and independent samples t-tests were conducted for gender, age and English proficiency at the pretest stage. The results showed no significant differences in gender distribution ($\chi^2 = 0.086$, $p = 0.769$), age ($t = -0.929$, $p = 0.357$) or English scores ($t = -1.715$, $p = 0.091$), indicating initial group comparability.

Before the intervention, all participants were informed of the study's purpose, procedures and data usage policies. Informed consent was obtained, and participation was entirely voluntary. All data were anonymous and used only for academic research, with no personally identifiable information collected. Importantly, to ensure fairness and mitigate potential educational equality concerns, the control group received equivalent instruction in the use of LLMs to support writing after the completion of the study.

3.3. Experimental procedures

This study employed a mixed-methods design combining quantitative procedures with qualitative interviews. The participants were two intact classes of first-year non-English majors. Both groups were taught by experienced instructors using *New World Interactive English (Second Edition)* as the main teaching material, with two 90-minute writing sessions held every 2 weeks. To minimize teacher effects, comprehensive coordination protocols were implemented throughout the intervention period. The two teachers collaborated to develop shared lesson plans for each writing task, ensuring identical topic selection, writing stages (planning, drafting, revising), and instructional timing. They also employed standardized scoring rubrics based on College English Test Band 4 (CET-4, a nationwide English proficiency examination that Chinese university students are typically required to take during their undergraduate studies) across both classes to maintain consistency in assessment. When schedules permitted, teachers conducted mutual classroom observations, with each observing the other's instruction at least once during the intervention. Additionally, they discussed teaching methods for each instructional task and addressed procedural issues. Any inconsistencies were discussed and standardized to maintain experimental integrity.

The intervention lasted for 12 weeks, covering the main instructional stages of the entire semester. The research timeline included a pretest administered in Week 1 (T1), followed by four biweekly writing tasks beginning in Week 3. A post-test was conducted in Week 12 (T2). The overview of the research procedure is shown in Figure 1. In the first writing task (T1), students in both groups completed an argumentative writing task on the same topic without any external intervention. Baseline data were also collected using an online questionnaire platform (www.wjx.cn), including demographic items and WSSRLQ.

The following four writing cycles were conducted under structured instructional conditions. The instructional design was adapted from Bai (2015) and focused on guiding students to use SRL strategies.

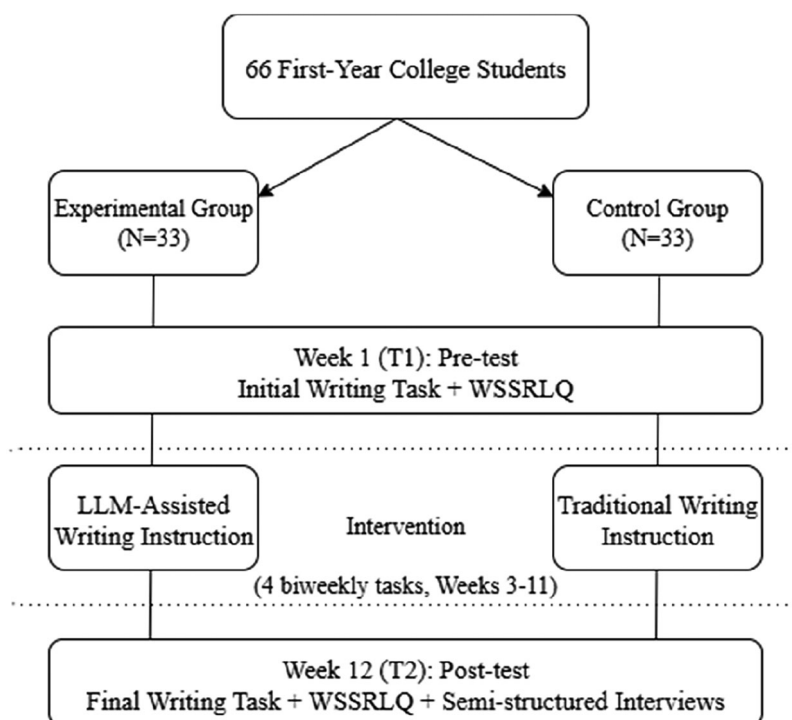


Figure 1. Experimental procedure.

Each task was divided into three stages: planning, drafting and revising. During the planning stage, students were guided to set goals and generate ideas (metacognitive strategies). In the experimental group, ChatGPT-4o was used to assist with idea generation, while the control group relied on teacher demonstration and peer discussion. During the drafting stage, students in both groups completed essays independently without external assistance, focusing on applying cognitive (e.g. text processing), meta-cognitive (e.g. self-monitoring) and motivational regulation strategies (e.g. motivational self-talk [MST]). In the revising phase, students were guided to apply social behavioral strategies by engaging with feedback and revising accordingly. The experimental group utilized ChatGPT-4o for automated feedback based on the CET-4 rubric covering language, grammar, coherence, and structure; the control group received teacher feedback using the same criteria.

In Week 12, after the final writing task, all participants completed the post-test WSSRLQ. Additionally, 11 students from the experimental group were invited to participate in semi-structured interviews to further explore their experiences with LLM-supported writing.

The specific prompts employed are outlined below. During the planning stage, students were encouraged to formulate more specific prompts to elicit structured and diverse outputs rather than relying on single, open-ended prompts. For initial brainstorming, they used prompts designed to generate multiple arguments. One example was: 'I am writing an argumentative essay on (specific topic). Please list 3–5 key arguments related to the topic. Provide at least one supporting example for each argument'. To broaden perspectives, students used prompts like: 'Analyze this topic from pedagogical, social, and technological perspectives to identify overlooked dimensions'. To build essay structure, they applied outline scaffolding prompts such as: 'Based on my initial ideas, please help me create a detailed outline on this topic, including the problems, causes, effects, and solutions'. Students were required to engage in iterative interaction with ChatGPT, typically exchanging at least two to three rounds of communication to refine their outlines, explore additional evidence, or clarify the scope of their arguments. They were also encouraged to request subject-specific vocabulary lists and useful expressions at the CET-4 level to enrich their language usage. A minimum of two conversational turns was required before drafting started.

After completing the draft, students used standardized rubric-based prompts aligned with CET-4 assessment criteria. A typical prompt was:

Please review this essay according to the CET-4 essay scoring standards. Provide scores out of 15 and detailed feedback for Language Accuracy and Vocabulary, Coherence and Logic, Structure and Organization, and Content Development. For each criterion, identify three specific improvement areas and provide concrete revision suggestions with examples.

During the revision stage, these structured prompts were provided as examples to help students focus on specific aspects of their writing. Students then made substantial revisions based on ChatGPT's feedback, including paragraph restructuring and logical refinement. Simultaneously, they were encouraged to engage in multiple rounds of interaction with ChatGPT, seeking explanations for specific suggestions and requesting additional guidance on potential improvements. This interaction with ChatGPT-4o aimed to cultivate students' skills in effectively using prompts and critically evaluating LLM outputs.

3.4. Instruments

3.4.1. Questionnaire

The questionnaire consisted of two parts. The first part collected participants' demographic information, including age, gender, academic major and English proficiency. The second part employed the WSSRLQ developed by Teng and Zhang (2016), drawing on Zimmerman (1989) framework and the core concepts of social cognitive theory. It was designed to assess learners' use of multidimensional WSSRL in the writing process.

The original WSSRLQ comprises four dimensions: cognitive, metacognitive social behavioral, and motivational regulation strategies, covering nine sub-strategies. The questionnaire uses a 7-point Likert scale ranging from 1 ('strongly disagree') to 7 ('strongly agree'). For the purpose of this study, course memory and peer learning dimension were excluded. As this study focused on real-time strategies during writing, the course memory dimension was excluded. In the original scale, this dimension mainly assessed recall and reuse of previously learned materials, which did not directly align with the objectives of the LLM-assisted teaching protocol. Peer learning was removed due to the individualized nature of the ChatGPT-4o intervention, which did not involve peer interaction. After a pilot test, items with low item-total correlations (<0.30), such as 'When writing, I tell myself to follow my plan', were removed to improve internal consistency. The final version of the questionnaire contained 31 items across seven WSSRL dimensions. It demonstrated high internal consistency (Cronbach's $\alpha = 0.942$). Suitability for factor analysis was confirmed (KMO = 0.887; Bartlett's test: $\chi^2 = 4774.80$, $p < 0.001$). Principal component analysis with varimax rotation extracted seven factors explaining 79.19% of the total variance, with all factor loadings above 0.58. Confirmatory factor analysis (using AMOS 24.0) indicated acceptable model fit indices ($\chi^2/df = 2.06$, RMSEA = 0.079, TLI = 0.908, CFI = 0.920), confirming the scale's construct validity and reliability for use in this study.

3.4.2. Semi-structured interview

To better understand students' use of WSSRL within the LLM-supported environment, semi-structured interviews were conducted with 11 students from the experimental group. Participants were purposefully selected based on their responses to a preliminary survey, aiming to capture diverse attitudes toward LLM integration. The interviews focused on two key areas: students' perceptions of LLM-supported strategy use (e.g. 'Has the use of LLMs changed the way you generate ideas?') and comparative evaluations of LLM-generated and teacher-provided feedback (e.g. 'How do you view the reliability of feedback from LLMs compared with teachers?').

Interviews were conducted in Chinese to ensure clarity of expression, lasted approximately 25 min, and were audio-recorded with informed consent. Professional transcription was completed by the first author, a native Chinese speaker with advanced English proficiency. Key excerpts underwent rigorous translation validation through back-translation by an independent bilingual researcher with applied linguistics expertise, followed by member-checking with participants to verify accuracy and preserve intended meanings. All discrepancies were resolved through collaborative discussion until consensus was achieved. Transcripts were anonymized for analysis.

3.5. Statistical analysis

Quantitative data were analyzed using SPSS version 27 (SPSS Inc., Chicago, IL). Descriptive statistics summarized WSSRL use at T1 and T2, and baseline equivalence was confirmed through group comparisons at T1. Normality was assessed via the Shapiro–Wilk test. Depending on distribution, paired t-tests or Wilcoxon signed-rank tests were used for within-group comparisons, and independent-samples t-tests or Mann–Whitney U tests for between-group comparisons. For variables showing significant baseline differences, ANCOVA was employed to control for initial disparities.

Qualitative data from 5.5 h of interviews (approximately 58,000 characters after cleaning) were analyzed using NVivo 20. Guided by the four WSSRL dimensions (cognitive, metacognitive, social behavioral, and motivational regulation strategies), a three-level thematic coding process was adopted. Initial coding was carried out at the semantic level. For example, the statement ‘When I have no ideas, I send a prompt to ChatGPT, and it guides me to think. Then I add my own ideas on this basis’ was first marked as ‘LLM-assisted idea generation’ (Level 1: initial code), then classified as ‘idea planning’ (Level 2: sub-theme), and finally categorized as ‘metacognitive strategies’ (Level 3: major SRL strategies). This bottom-up coding process both retained the details in the data and maintained the corresponding relationship with the WSSRL model. Finally, a total of 398 initial codes were identified, and these were further grouped into 9 subtopics. Seven of them corresponded to the existing model, while the other two emerged inductively from the data, namely ‘self-initiated feedback seeking’ and ‘critical engagement with feedback’, and were classified under a new category: LLM-mediated feedback strategies, reflecting learners’ adaptive use of LLM-generated feedback. Inter-coder reliability was confirmed by a second trained coder (Cohen’s $\kappa = 0.823$), indicating substantial agreement.

4. Results

4.1. Quantitative results

4.1.1. Baseline equivalence

To ensure comparability between groups prior to the intervention, baseline analyses were conducted at T1. As shown in Table 2, there were no significant differences that emerged in five of the seven WSSRL strategies: text processing (TP), idea planning (IP), feedback handling (FH), MST and emotional control (EC), indicating general equivalence between groups. However, the control group scored significantly higher than the experimental group in goal-oriented monitoring and evaluation (GME) and interest enhancement (IE). These baseline differences were statistically controlled using ANCOVA in subsequent analyses.

4.1.2. Within-group comparisons

Within-group comparisons revealed clear differences in WSSRL development in two groups (see Table 3). The experimental group demonstrated noticeable increases across six WSSRL, including TP ($Z = -2.796$, $p < 0.01$), IP ($Z = -3.625$, $p < 0.01$), GME ($t = -4.731$, $p < 0.01$), IE ($t = -4.321$, $p < 0.01$), MST ($Z = -2.893$, $p < 0.01$) and EC ($t = -2.948$, $p < 0.01$). The only exception was FH, which showed a significant decline from $M = 4.05$ ($SD = 0.75$) at T1 to $M = 3.76$ ($SD = 0.64$) at T2 ($Z = -2.021$, $p < 0.05$).

Table 2. Between-group comparison of WSSRL.

Category	Scale	T1				T2			
		EG M(SD)	CG M(SD)	t / U	p	EG M(SD)	CG M(SD)	t / U	p
Cognition	TP	3.06 (0.82)	3.37(0.52)	417.5	0.102	3.53(0.36)	3.40(0.42)	1.298	0.199
Metacognition	IP	2.80(1.13)	2.66(0.46)	478.5	0.391	3.67(0.68)	2.78(0.53)	5.899	0.000**
	GME	2.83 (0.81)	3.31(0.86)	−2.340	0.022*				
Social behavioral	FH	4.05 (0.75)	4.19(0.64)	498.0	0.549	3.76(0.64)	4.17(0.58)	347.0	0.011*
Motivational regulation	IE	2.89 (0.96)	3.72(0.74)	−3.965	0.000**				
	MST	3.02 (0.75)	3.32(0.94)	−1.268	0.209	3.51(0.45)	3.29(0.73)	469.0	0.331
	EC	3.20 (0.81)	3.64(0.84)	432.0	0.145	3.63(0.52)	3.51(0.53)	0.952	0.345

* $p < 0.05$.

** $p < 0.01$.

Table 3. Within-group comparison of WSSRL.

Category	Scale	Group	T1 M(SD)	T2 M(SD)	t / Z	p
Cognition	TP	EG	3.06 (0.82)	3.53 (0.36)	−2.796	0.005**
		CG	3.37 (0.52)	3.40 (0.42)	−0.969	0.333
Metacognition	IP	EG	2.80 (1.13)	3.67 (0.68)	−3.625	0.000**
		CG	2.66 (0.46)	2.78 (0.53)	−2.359	0.018*
	GME	EG	2.83 (0.81)	3.51 (0.32)	−4.731	0.000**
		CG	3.31 (0.86)	3.42 (0.81)	−0.929	0.360
Social behavioral	FH	EG	4.05 (0.75)	3.76 (0.64)	−2.021	0.043*
		CG	4.19 (0.64)	4.17 (0.58)	−0.185	0.853
Motivational regulation	IE	EG	2.89 (0.96)	3.66 (0.68)	−4.321	0.000**
		CG	3.72 (0.74)	3.43 (0.42)	2.016	0.052
	MST	EG	3.02 (0.75)	3.51 (0.45)	−2.893	0.004**
		CG	3.28 (0.91)	3.29 (0.73)	−0.048	0.962
	EC	EG	3.20 (0.81)	3.63 (0.52)	−2.948	0.006**
		CG	3.39 (0.65)	3.51 (0.53)	−1.158	0.247

* $p < 0.05$.** $p < 0.01$.**Table 4.** ANCOVA results for GME and IE between groups at T2.

Category	Scale	Group	EG M(SD)	CG M(SD)	Adjusted mean	SE	F	η^2	p
Metacognition	GME	EG	2.83 (0.81)	3.51 (0.32)	3.602	0.094	4.145	0.062	0.046*
		CG	3.31 (0.86)	3.42 (0.81)	3.325	0.094			
Motivational regulation	IE	EG	2.89 (0.96)	3.66 (0.68)	3.712	0.103	4.601	0.068	0.046*
		CG	3.72 (0.74)	3.43 (0.42)	3.382	0.103			

* $p < 0.05$.** $p < 0.01$.

In contrast, most WSSRL scores in the control group remained stable between T1 and T2. The only significant change was in IP, which increased modestly from $M = 2.66$ ($SD = 0.46$) to $M = 2.78$ ($SD = 0.53$) ($Z = -2.359$, $p < 0.05$). The lack of significant differences in other dimensions suggests that the control group's WSSRL use was consistent throughout the semester without LLM support.

4.1.3. Between-group comparisons

Between-group comparisons at T2 (Table 2) revealed significant differences between groups in IP ($t = 5.899$, $p < 0.001$) and FH ($U = 347.0$, $p < 0.05$). The experimental group outperformed the control group in IP while scoring lower in FH. ANCOVA was performed to control for baseline differences in GME and IE (see Table 4). After adjustment, both GME ($F = 4.145$, $p < 0.05$) and IE ($F = 4.601$, $p < 0.05$) scores were significantly higher in the experimental group, supporting the positive effect of LLM-assisted instruction.

4.2. Qualitative results

Semi-structured interviews were conducted with 11 participants from the experimental group to explore LLM impact on their WSSRL use. Interview analysis revealed that students applied various strategies in the LLM-supported writing process.

4.2.1. Self-regulated writing strategies within the established SRL framework

Interview data were first analyzed using the four WSSRL dimensions: cognitive, metacognitive, social behavioral and motivational regulation.

In the cognitive domain, students reported a wide improvement in text processing. LLM helped enhance lexical variety, sentence complexity, and grammar accuracy. S1 pointed out: 'ChatGPT can optimize my sentence structure and make it more complex and diverse'. S9 added, 'I've been studying grammar hard since high school. ChatGPT helps me identify writing mistakes, which is very beneficial'.

In the use of metacognitive strategies, students noted gains in idea planning and goal-setting. For idea planning, they generally expressed that LLM helped outline generation and clarified task requirements. S3 remarked, 'The outline of AI is more structured than mine. That helps me brainstorm'. S4

echoed this view, saying, 'When I feel stuck, the outline provides direction, and I build upon it with my ideas'. In GME, some students regarded LLM as an auxiliary tool for writing reflection. S1 explained, 'I try to understand why ChatGPT revised a sentence and then apply that knowledge to future writing'.

Social behavioral strategies centered on feedback preferences. Many students favored LLM feedback due to its immediacy and emotional neutrality. S3 noted, 'AI can partly replace teacher feedback, helping me rephrase repetitive words and easily correct grammar'. S7 stated, 'I am introverted and dislike face-to-face feedback. I prefer using ChatGPT because there is no social pressure'. These responses suggested that students' feedback handling was influenced not only by the content quality but also by emotional comfort in human-LLM interactions.

Motivational regulation also improved. The novel experience provided by LLM improved students' engagement. S8 said, 'Using ChatGPT feels fresh and makes the classroom more interesting'. For motivational self-talk, students set goals and recognized their progress. S7 shared: 'I try to accumulate more complex vocabulary and push myself to write better than before'. S10 mentioned: 'I'm more confident to express my ideas now'. Regarding EC, several students reported reduced anxiety due to LLM's clear and constructive suggestions. S2 pointed out: 'AI scoring shows where I need to improve. This is clearer than just getting a total score'.

4.2.2. Emergent LLM-mediated feedback strategies

Beyond the established WSSRL framework, interview data revealed two behavior patterns that were not fully covered by the original model, namely self-initiated feedback seeking and critical engagement with feedback. Together, these constituted a new dimension labeled LLMs-mediated feedback strategies.

In terms of self-initiated feedback seeking, many students actively requested multiple rounds of feedback from ChatGPT, especially during revision. Its immediacy, accessibility and ability to provide specific suggestions were widely recognized. S3 shared, 'ChatGPT can well understand my learning needs and give immediate feedback'. S4 emphasized the change of feedback rhythm: 'I used to spend a week rewriting an article, but now I can finish it in one day'. In addition, LLM's non-judgmental nature provided psychological comfort. S7 said, 'It's more comfortable asking ChatGPT than requesting face-to-face teacher feedback'. Students reported that LLM could identify grammatical and lexical issues precisely, which made the feedback process more efficient and less dependent on teacher availability.

Despite their appreciation for technical support, many students also exercised judgment in handling LLM-generated feedback. Some questioned the appropriateness of certain suggestions, especially when LLM over-revised text or assigned inconsistent scores. S1 remarked, 'It replaces all my phrases with fancy words that I can't even use correctly'. S9 questioned scoring consistency: 'The total score didn't change between drafts, even when the sub-scores did'. When facing LLM feedback uncertainty, students actively sought clarification and supplementary suggestions from teachers. S4 said, 'My teacher can provide more targeted advice. I think communicating with her is very meaningful'. These responses suggested that students were developing feedback literacy, using LLM suggestions strategically while realizing the value of teacher feedback.

5. Discussion

5.1. Enhancing WSSRL use through LLM-assisted writing instruction

Quantitative and qualitative results showed that the experimental group students made significant progress in cognitive, metacognitive and motivational regulation strategies compared to the control group. These findings reveal that embedding LLMs into structured writing instruction provides effective scaffolding that fosters deeper and more strategic learner engagement.

At the cognitive level, experimental group participants showed obvious improvement in text processing strategies. With LLM support, students could identify and correct grammatical errors more effectively, enhance syntactic diversity and enhance the clarity of expression. Through technology-mediated language input, LLMs enabled students to observe and internalize suggestions regarding vocabulary use, sentence structure and academic expression (Zhao et al., 2023). Previous studies have confirmed that LLMs could identify common errors in grammar and punctuation while providing specific modification

guidance (Marzuki et al., 2023). Therefore, they play a positive role in optimizing linguistic quality (Wale & Kassahun, 2024).

In the metacognitive dimension, students made great progress in idea planning and GME. This indicated that introducing LLMs into writing instruction helps students better organize ideas before writing and actively reflect on their writing process. Many participants mentioned that utilizing ChatGPT to generate outlines and supplementary points enabled them to grasp the writing direction more clearly. Lingard (2023) noted that ChatGPT could provide multiple outline versions and enhance writing depth by presenting counterarguments or supplementing theoretical backgrounds. Wang (2025) found that ChatGPT assisted learners in brainstorming and text framework construction, helping optimize structure and stimulate idea generation. Such support reduced students' anxiety and improved writing confidence. In addition, students also actively used LLM-generated feedback for writing process reflection. Some learners drew on suggestions to determine whether their drafts met expected goals or to revise the text based on logic and clarity. Wale and Kassahun (2024) noted that AI tools could help students track writing changes over time and promote more systematic revisions.

In terms of motivational regulation, students exhibited increased use of IE, motivational self-talk and EC strategies. Technology novelty and feedback immediacy appeared to improve writing participation and self-confidence. Many learners began to regard writing as a continuously improvable skill and maintained writing motivation through self-encouraging language. This aligned with Özçelik and Ekşi (2024) research, which argued that ChatGPT can significantly enhance learners' interest and engagement, particularly among those with low writing confidence. Wang (2025) further asserted that the integration of LLMs may foster learners' self-efficacy and mitigate writing-related anxiety. Ratih and Kastuhandani (2024) found that instant feedback helped learners see their progress more clearly and also maintained their positive emotions to a certain extent.

Notably, regarding the social behavior strategy, the frequency of feedback handling strategy used by the experimental group decreased. However, qualitative results revealed that this change did not mean that students ignored feedback but reflected a transition in feedback sources. Compared with traditional teacher or peer feedback, they relied mainly on LLM-provided feedback. This tendency is related to the availability, efficiency and low-pressure characteristics of AI feedback.

Previous studies have shown that LLMs are changing academic support in higher education. For example, ChatGPT can provide real-time responses, personalized suggestions and continuous feedback, bringing unprecedented flexibility to writing guidance and revision (Chan & Colloton, 2024). Onesio-Ozigagun et al. (2024) noted that LLMs transcend temporal and spatial constraints to create efficient feedback environments. With ChatGPT, students can receive timely, multi-level feedback on content, structure and language, which helps improve revision efficiency and self-regulation ability (Bašić et al., 2023; Wei & Li, 2023; Wu, Lee, et al., 2024; Wu, Li, et al., 2024). Additionally, compared with traditional face-to-face communication, LLMs' non-critical characteristics reduce psychological anxiety often associated with teacher-student communication (Deng, 2024). This low-pressure feedback environment allows students to seek clarification repeatedly without hesitation until their questions are fully resolved.

5.2. Emerging LLM-mediated writing strategies

Interview data revealed that students showed two emerging feedback-related behaviors beyond the traditional WSSRL framework: self-initiated feedback seeking and critical engagement with feedback.

The study observed that many students actively initiated questioning and requested LLMs to provide further explanations or operational guidance. This dynamic interaction relies on the real-time response and accessibility of LLMs, enabling students to complete multiple rounds of iterative correction within a relatively short time. Continuous interaction not only improves revision efficiency but also deepens understanding of language forms and text structure. Banihashem et al. (2024) highlighted that LLM feedback often met academic writing standards with low error rates. This feature encourages students to implement optimization strategies more systematically. Such behavior reflects individuals' ability to mobilize external resources, which constitutes a core element of SRL.

Research data also revealed that students did not blindly accept all LLM-provided suggestions. Instead, they made selective responses based on their writing purposes and expression intentions. When

students were uncertain or dissatisfied with LLM-generated outputs, they would critically evaluate and cross-check the information using alternative sources (Liu, Park, et al., 2024). Shi and Aryadoust (2024) found that learners were highly receptive to form-focused adjustments such as grammar and vocabulary, but had reservations about content-related feedback, which was closely related to current LLMs' limited contextual understanding. This prudent screening process reflects students' developing feedback literacy, with a decision-making process including dual considerations of quality assessment and goal matching (Tubino & Adachi, 2022). Rather than passively receiving suggestions, students actively evaluate the relevance and appropriateness of external input during feedback interactions.

The decline in traditional feedback handling strategies also highlights a key limitation of existing WSSRL measurement tools in LLM-assisted learning environments. The WSSRLQ conceptualizes feedback handling as a relatively static construct centered on interpreting inputs from human sources such as teachers and peers (Teng & Zhang, 2016). However, present findings suggest this framework fails to fully capture LLM feedback's dynamic and interactive nature. Rather than being passive recipients, students take active roles in interpreting, evaluating and applying AI feedback (Winstone et al., 2017). Such agency is particularly evident in their multiple rounds of interaction with ChatGPT during revision. Real-time formative feedback from LLMs enables students to rapidly identify strengths and weaknesses, providing 'mastery experiences' that enhance self-efficacy (Teng, 2024a, 2024b, 2024c). This leads them to submit drafts to ChatGPT before seeking human feedback. Consequently, the decline in FH scores is evidence of a fundamental shift in feedback sources and interaction patterns that existing scales struggle to capture. This highlights the need to update theoretical models and measurement tools to distinguish between different feedback sources and source-specific strategies. Notably, although students actively engaged with LLMs, teacher feedback remains irreplaceable in writing guidance. Compared with LLMs, teachers' pedagogical expertise and contextual awareness enable them to provide more targeted and developmental guidance. This aligns with Han and Li (2024) findings, which confirm that while ChatGPT can provide instant and standardized suggestions, it lacks teaching depth and adaptability. Similarly, Steiss et al. (2024) propose introducing LLM feedback during initial writing stages while stressing teachers' core value in improving language accuracy and maintaining pedagogical coherence. Overall, these findings highlight the complementary relationship between LLMs and teachers. The innovative teaching framework helps improve students' engagement in the feedback process and judgment abilities.

6. Conclusion

This study explored the impact of integrating ChatGPT into college English writing instruction on learners' WSSRL development. The findings indicate that LLM-assisted instruction effectively promotes students' cognitive engagement, metacognitive planning and motivational regulation in writing tasks. In addition, the emergence of self-initiated feedback seeking and critical engagement with feedback behaviors suggests that students are developing new forms of learning strategies in response to AI-mediated support. While students appreciated the efficiency and immediacy of LLM-generated feedback, they also expressed a consistent need for teacher input, especially in areas requiring contextual judgment and individualized guidance. The value of combining LLMs with teacher support to foster sustainable feedback literacy and learner autonomy is becoming increasingly evident.

This study offers some pedagogical implications. For teachers, effective integration of LLMs should support specific writing stages, such as planning or revising, while still providing personalized and context-aware guidance, especially in terms of genre norms and rhetorical choices. Teachers should also guide students to critically evaluate LLM-generated feedback and help them learn when and how to apply suggestions rather than blindly accepting them. For university administrators, it is necessary to establish standardized policies for the responsible use of LLMs, provide professional training for faculty, and ensure access to secure, stable digital platforms. For curriculum designers, incorporating LLMs requires not only aligning technology with learning goals and pedagogical models, but also rethinking assessment methods. Additionally, digital literacy development needs integration into writing instruction to help students better prepare for technology-supported learning environments.

While this study contributes valuable insights into LLM-assisted writing instruction, several methodological considerations point toward promising directions for future research. First, we relied solely on

self-reported strategy data without triangulating through objective assessments of writing performance. Self-report measures are susceptible to retrospective memory errors and participants' limited metacognitive awareness of their actual strategy use. Without systematic writing evaluations and language accuracy measurements, it is difficult to establish clear connections between reported strategy use and actual learning outcomes. Future research would benefit from mixed-method studies that combine objective performance measures with self-reported data, employing standardized scoring rubrics, language accuracy assessments and revision-tracking analyses to verify whether reported strategy shifts translate into measurable writing progress.

Second, the relatively homogeneous participant sample limits applicability to diverse learner populations with varying writing habits, learning styles and technology attitudes. All participants were Chinese EFL learners from a single university within specific disciplines, while learners across different majors may display distinct technological adaptation and learning preferences. Future investigations should expand to diverse populations and contexts to build a more comprehensive understanding of how various learners respond to LLM-assisted instruction.

Third, despite efforts to ensure instructional consistency through shared planning and observation, the quasi-experimental design with different instructors assigned to each group introduces potential confounding variables that could not be fully eliminated. Instructor effects, including teaching style, feedback quality and classroom management, remain uncontrolled variables that could account for observed differences. Methodological rigor demands randomized controlled trials with a single instructor to eliminate confounding variables related to teaching style and classroom dynamics, providing clearer evidence of intervention effectiveness while maintaining ecological validity.

Fourth, temporal constraints may be insufficient to reflect the long-term development of WSSRL. The 12-week intervention period, while broadly covering a full semester, may be inadequate to capture deep and enduring changes in strategic behavior. Strategy development typically requires extended periods, and the persistence of observed changes remains unclear. Longitudinal research designs extending beyond a single semester are crucial for distinguishing novelty effects from sustained pedagogical change. Such studies should incorporate multiple follow-up assessments to evaluate the persistence of strategy development and the sustainability of initial gains in strategy use. By addressing these considerations through more comprehensive and rigorous research designs, future studies can more effectively guide educators in integrating LLM-assisted instruction into their teaching practices, thereby enhancing students' strategy use and learning performance.

CRedit authorship contribution statement

We confirm that the manuscript has been read and approved by all named author and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

All authors are as follows:

Aoran Zhang (First Author): Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Software; Visualization; Writing – Original Draft; Writing – Review & Editing.

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The datasets generated during and/or analyzed during this study are available from the corresponding author on reasonable request.

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